

Preface

If we knew what it was we were doing, it would not be called research, would it?

Albert Einstein

The Science of Algorithmic Trading and Portfolio Management is a reference book intended to provide traders, portfolio managers, analysts, students, practitioners, and financial executives with an overview of the electronic trading environment, and insight into how algorithms can be utilized to improve execution quality and fund performance.

We provide a discussion of the current state of the market and advanced modeling techniques for trading algorithms, stock selection, and portfolio construction.

This reference book will provide readers with:

- An understanding of the new electronic trading environment.
- An understanding of transaction cost analysis (TCA) and proper metrics for cost measurement and performance evaluation.
- A thorough understanding of the different types of trading algorithms: liquidity seeking, dark pools, arrival price, implementation shortfall (IS), volume weighted average price (VWAP), arrival price, and portfolio implementation shortfall.
- Proven market impact modeling techniques.
- An understanding of algorithmic trading across various asset classes: equities, futures, fixed income, foreign exchange, and commodities.
- Advanced algorithmic forecasting techniques to estimate daily liquidity and monthly volumes.
- An algorithmic decision making framework to ensure consistency between investment and trading objectives.
- A best execution process.

Readers will subsequently be prepared to:

- Develop real-time trading algorithms customized to specific institutional needs.
- Design systems to manage algorithmic risk and dark pool uncertainty.
- Evaluate market impact models and assess performance across algorithms, traders, and brokers.
- Implement electronic trading systems.

For the first time, portfolio managers are not forgotten and will be provided with proven techniques to better construct portfolios through:

- Stock Selection
- Portfolio Optimization
- Asset Allocation
- MI Factor Scores